

Donkeys, bishops, and the distinction between weak and strong definites

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Abstract

Much recent work on definiteness has assumed that definite descriptions come in two types, weak and strong. Weak definites make use of a simple situation-relative uniqueness operator; strong definites add to this a variable available to dynamic binding. Drawing on evidence from Mandarin, Jenks (2018) influentially proposed that strong definites but not weak definites can serve as donkey anaphors. We show that this claim is incorrect for Mandarin as well as for German and English. Certain special cases of donkey anaphora nevertheless have diagnostic value for the distinction between weak and strong definites. We discuss two. First, in structures where there is no dynamic quantifier available to serve as an antecedent, strong definite descriptions cannot serve as donkey anaphors. Second, in so-called bishop sentences, the uniqueness requirement of the weak definite is not met; in these cases weak definite descriptions cannot serve as donkey anaphors. We apply these tests to definites in German, English, and Mandarin. Our results indicate that English *the* and Mandarin bare nouns are always weak definites, but also that important aspects of crosslinguistic variation with respect to weak definites remain to be explained.

1 Introduction

Schwarz (2009, 2013) influentially proposed that definite descriptions come in two distinct types. Weak definites require uniqueness of the referent relative to the NP description and a situation. Strong definites add a referential index available to dynamic binding: they require uniqueness relative to the NP description, a situation, and the value of the index as determined by the assignment function. In German, Schwarz argued, definite articles are disambiguated when they occur as complements of prepositions. Weak definites contract with prepositions (e.g. *von* ‘of’ + DEF_{weak} = *vom*) and strong definites do not.

(1) Weak definite (e.g. *vom Senator* ‘of the senator’)

a. [PP P [DP DEF_{weak} NP]]

(Schwarz 2009: 265)

$$b. \llbracket \text{DEF}_{\text{weak}} \rrbracket^g = \lambda s. \lambda P. \iota x [P(x)(s)] \quad (\text{Schwarz 2009: 264})$$

(2) Strong definite (e.g. *von dem Senator* ‘of the senator’)

$$a. [\text{PP } P [\text{DP } 1 [\text{DEF}_{\text{strong}} \text{ NP}]]] \quad (\text{Schwarz 2009: 265})$$

$$b. \llbracket \text{DEF}_{\text{strong}} \rrbracket^g = \lambda s. \lambda P. \lambda y. \iota x [P(x)(s) \wedge x = y] \quad (\text{Schwarz 2009: 264})$$

The idea that these two distinct types of definiteness co-exist is often introduced (including by Schwarz) with reference to two competing intuitions about definiteness: that it is fundamentally about uniqueness, or that it is fundamentally about familiarity. Each intuition gives rise to a particular technology. For the uniqueness theory, as (1) shows, the relevant technology is situation-relative uniqueness, along with the broader machinery of situation semantics needed to cash this out (e.g. Heim 1990, Elbourne 2001, 2005). For the familiarity theory, as in (2), it is the machinery of individual variables and the binding theory needed to properly manipulate them (e.g. Heim 1982, Groenendijk and Stokhof 1991).

In setting up this formulation, we take pains to distinguish the starting intuition from the resulting technology. A technology, as part of a linguistic theory, can be tested; an intuition cannot. In the wake of Schwarz’s work, a substantial literature has emerged finding evidence for both technologies in the definite expressions of additional languages, including Mandarin (Jenks 2018), Korean (Ahn 2017), Thai (Jenks 2015, Moroney 2021b), Chuj (Royer 2022a,b), A’ingae (Zheng and AnderBois 2023), Wu (Simpson 2017), and Marka-Dafing (Jenks and Konate 2022), as well as pronouns in a variety of languages (Patel-Grosz and Grosz 2017, Clem 2017, Bi 2018, Bi and Jenks 2019, i.a.). What this literature overall shows most directly is that both types of technology are needed.¹ As Coppock (2022) emphasizes, this is not because the technology of uniqueness cannot account for basic data that *intuitively* support familiarity (it can) nor that the technology of familiarity cannot account for basic data that *intuitively* support uniqueness (it can as well). Rather, the argument is that there are certain relatively specific corners of the usage of definites where uniqueness- and familiarity-based technologies make different predictions, and languages make use of definite forms in distinctive ways in these contexts. Some definite forms are revealed to be specialized for uniqueness uses, whereas others are specialized for familiarity.

Our central interest in this paper is the proper characterization of these corner cases. We focus especially on potential test environments that involve donkey anaphora, i.e. co-variation between a definite expression and a non-c-commanding antecedent. (Note that we use the term ‘definite expression’ in a broad way, including demonstratives.) The three classes of examples on which we will focus are shown for English in (3).²

- (3) a. Most farmers who bought a donkey treated **the/that donkey** well.
 b. Every dog-owner in this building already suspects that **the/#that dog** is bothering the neighbors.

¹In this it builds on an earlier literature defending both dynamic and non-dynamic components to anaphora, including Kratzer (1989), Chierchia (1992), Cheng and Huang (1996).

²The judgments indicated here are for the co-varying (donkey) reading; (3b) is well-formed with *that* on an exophoric reading.

- c. If an employee is bothering other employees, we must fire **that/#the employee**.

A claim made relatively frequently in the literature is that the weak/strong distinction regulates which definite forms may be used as donkey anaphors in examples like (3a): only strong definites can be donkey anaphors (Jenks 2015, 2018, Royer 2020, 2022a,b, Moroney 2021a,b, Sharma 2025; Dawson and Jenks 2023 concur with slight hedging). We will show that this claim is incorrect; in fact Schwarz (2009) has already shown for German that it is incorrect. Rather, we contend, the proper types of donkey sentences to consider are those in (3b) and (3c). Each of these sentences manipulates a factor central to one of the definiteness technologies. Sentence (3b) exemplifies what we call a *no proper binder* structure: it lacks a full indefinite antecedent for the donkey definite. Without a proper antecedent, a strong definite in the main clause cannot be dynamically bound. This rules out the strong definite; only the weak definite survives. Sentence (3c) exemplifies a *bishop sentence*: there is not a unique individual satisfying the descriptive content of the definite. This rules out the weak definite; only the strong definite survives.

Our secondary interest in this paper is the application of the weak/strong definiteness distinction to English and to Mandarin. In the wake of Schwarz 2009, most recent work has assumed that English *the* is weak/strong ambiguous. We argue instead for a view suggested by Schwarz 2013: the German weak/strong distinction parallels the English difference between *the* and *that*. *The* is only a weak (uniqueness) definite. The strong (indexed) definite in English requires *that*. For Mandarin, the weak/strong distinction has engendered significant recent debate (Jenks 2018, Bremmers et al. 2022, Dayal and Jiang 2022, Simpson and Wu 2022, Saha et al. 2024, Jenks and Chen 2026), primarily focused on anaphora in non-donkey structures; both uniqueness and familiarity technologies can apply here (Coppock 2022). We show that no-proper-binder and bishop sentences are useful in this case. The picture we arrive at is one where weak definites are expressed with bare nouns in Mandarin, and strong definites with demonstratives, following Jenks (2018).³

In the next section, we present our case that basic donkey anaphora examples do not diagnose weak vs. strong definiteness, focusing largely on Mandarin. We then turn to examples of the type in (3b) and (3c) in §3 and §4, respectively, discussing evidence from German, English, and Mandarin. We conclude in section 5 with a brief look at outstanding questions of variation in this three-language sample, which lead us to suggest variation internal to the classes of uniqueness definites and familiarity definites across languages.

2 Basic cases of donkey anaphora

We begin with the argument that simple cases of donkey anaphora do not diagnose weak vs. strong definiteness. So far as we know, the origin of the claim that they do lies in

³This perspective is fully compatible with the conclusion in much recent work that definites do not compete with each other in the way that Jenks envisioned (Simpson and Wu 2022, Bremmers et al. 2022, Dayal and Jiang 2022, Zheng and AnderBois 2023, Ahn and Zhu 2025). We will see that both definite forms are possible in various Mandarin examples, providing additional support for non-competition views.

Jenks 2015, 2018, esp. Jenks' (2018) discussion of the Mandarin examples in (4). Here (4a) is a standard relative-clause type donkey sentence; (4b) is a *dou*-conditional (Cheng and Huang 1996). In both examples, the covarying reading is naturally expressed with the demonstrative, not the bare noun. Jenks argues that the demonstrative is a strong definite whereas the bare noun is a weak definite (a position we will ultimately agree with). He concludes from (4) that only strong definites can be donkey anaphors in Mandarin.⁴

- (4) a. mei ge you yi zhi shuiniu de nongfu dou hui da **#(na zhi) shuiniu**
 every CLF have one CLF buffalo REL farmer all will hit DEM CLF buffalo
 'Every farmer that has a buffalo hits that buffalo.' (Jenks 2018: 512)
- b. ni jiao shei jinlai, wo dou jian **#(na ge) ren**
 2SG ask who enter 1SG all see DEM CLF person
 'Whoever you ask to come in, I will see that person.' (Jenks 2018: 511, following Cheng and Huang 1996: 130)

While we do not contest the judgments in (4), we note that a broader range of examples makes Jenks' generalization about donkey anaphora in Mandarin untenable. The examples in (5)-(6) are like (4a) in various structural properties; in particular, the antecedent inside the relative clause is expressed in a parallel way (with *yi* 'one' and a classifier). Yet in these examples, both the demonstrative and the bare noun support a co-varying reading.⁵

- (5) mei ge na-guo yi mei aoyun jiangpai de yundongyuan dou ba **(na mei) jiangpai** haohao zhencang qilai-le
 every CLF take-EXP one CLF Olympics medal REL athlete DOU BA DEM
 CLF medal well cherish RES-ASP
 'Every athlete who has won an Olympic medal cherished the/that medal well.'
- (6) mei ge you yi ming latuoweiya qiuyuan de bingqiu dui dou hui rang
 every CLF have one CLF Latvian player REL hockey team DOU will let
(na ming) latuoweiya qiuyuan dang shoumenyuan
 DEM CLF Latvian player serve.as goalie
 'Every hockey team that has one Latvian player would make the/that Latvian player the goalie'

In view of these data, we conclude that the mere presence of donkey anaphora is not enough to require the use of a demonstrative in Mandarin.⁶ Some additional factor or factors must

⁴The following abbreviations are used in Mandarin glosses: ASP = aspect, CLF = classifier, COMPL = completive, DEM = demonstrative, EXP = experiential, REL = relative marker, RES = resultative.

⁵Uncited Mandarin examples were constructed by the second author and verified with 5-7 additional native speakers, ages 23-31, from Guangxi, Heilongjiang, Jiangsu, Shandong and Zhejiang.

⁶On this point we concur with Simpson and Wu (2022) (who likewise object to Jenks' generalization). Bare-noun donkey sentences have also been discussed in work addressing other issues in Mandarin donkey anaphora, e.g. Lin et al. (2020):

be in the mix. In principle, a number of different factors could be responsible for the difference between these examples and Jenks' relative clause donkey example (4a), including pragmatic inferences, parsing expectations, or syntactic or semantic differences orthogonal to donkey anaphora—all of which are explored in the broader literature on the choice between bare nouns and demonstrative DPs in Mandarin (Bremmers et al. 2022, Dayal and Jiang 2022, Simpson and Wu 2022).⁷ Plausibly, a mix of these factors is at play in (4a) in a way that relates to aspect/genericity. In (4a), given world knowledge, a reading where the object in the main clause is read generically is extremely plausible: a farmer that has a buffalo is a buffalo-hitter. This is the reading that the bare noun version of (4a) prominently receives, as has been noted before (e.g. Bi 2018). Perhaps the overwhelming plausibility of this reading makes it hard to perceive any other reading for the bare noun version, e.g. the donkey-anaphoric version. As expected, when we minimally alter the sentence to make a generic reading less plausible, a donkey reading of the bare noun becomes possible, as shown in (7).

- (7) mei ge you yi zhi shuiniu de nongfu zuihou dou sha-le (na zhi)
 every CLF have one CLF buffalo REL farmer eventually DOU kill-ASP DEM CLF
shuiniu.
 buffalo
 'Every farmer who has a buffalo eventually kills that buffalo.'

These data of course leave open various interesting questions about when demonstratives vs. bare nouns are used in Mandarin (a quite complex matter, as existing literature has amply shown). We conclude, *contra* Jenks (2018), that a very simple view of how this choice is made in donkey contexts cannot be maintained. Donkey anaphora does not require the use of a demonstrative in Mandarin.

Similar points can be made for conditional-type donkey sentences. A minor modification of the main clause in (4b) is sufficient to allow a bare noun to serve as a donkey anaphor, (8). Example (9) shows the same result can be replicated in an overtly 'unconditional' structure (cp. Rawlins 2013) and with a different choice of *wh*-word.

- (8) ni jiao shei jinlai, wo dou hui gen (na ge) ren dihui ni
 2SG ask who enter 1SG DOU will with DEM CLF person disparage 2SG
 'Whoever you ask to come in, I will disparage you in front of that person.'

- (i) mei ge you qiqiu de xiaopengyou dou xihuan **qiqiu.**
 every CLF have balloon REL child DOU loves balloon.
 Every child that has (a) balloon(s) loves the balloon(s). (Lin et al. 2020)

Our observations are also similar to those of Bi (2018), who notes that the pronominal version of (4a) behaves unlike other pronominal donkey anaphora examples.

⁷Simpson and Wu (2022) have noted for instance that the *ba*-construction typically allows both bare nouns and demonstratives to be anaphoric (a potentially syntactic influence); the *ba*-construction similarly allows both bare nouns and demonstratives to serve as donkey anaphors, as shown in (5).

- (9) bulun ni zuihou mai na ping jiu, wo dou hui ba (**na ping**)
 no.matter 2SG eventually buy which CLF wine 1SG DOU will BA DEM CLF
jiu he wan
 wine drink COMPL
 ‘No matter which wine you end up buying, I will finish drinking that wine.’

Here, too, there is no support for a ban on bare nouns as donkey anaphors in Mandarin.

Our conclusions mean that two claims from Jenks 2018 are not co-tenable. One, Mandarin demonstratives are strong definites and bare nouns are weak definites. Two, donkey anaphora requires strong definites. We note that evidence beyond Mandarin already suggests that the second of these claims cannot be correct. In particular, while Schwarz (2009) discusses some German examples where strong definites are needed for covarying readings, he also shows that German weak definites allow covarying readings:

- (10) Jeder Mann, der ein Haus mit Garten gekauft hat und die meiste Zeit zu Hause
 Every man that a house with yard bought has and the most time at home
 verbringt, arbeitet viel { **im** / **in dem** } **Garten**
 spends works much in.the.WEAK in the.STRONG yard
 ‘Every man who bought a house with a yard and spends most of his time at home
 works a lot in the yard.’ (Schwarz 2009: 45)

Notable in (10) is the possibility of a covarying reading not just for the strong definite *dem Garten* but for the weak definite *im Garten* as well. In the broader literature, evidence that weak definites can be donkey definites is available for Dafing (Jenks and Konate 2022: 27), A’ingae (Zheng and AnderBois 2023), and perhaps Guébie (Agodio et al. 2024).^{8 9}

In logical terms, rejecting the view that donkey anaphora requires strong definites still leaves it open whether Mandarin in fact distinguishes weak from strong definites using bare-noun vs. demonstrative DPs, as per Jenks 2018. The literature is not in agreement on this point. Dayal and Jiang (2022), for instance, propose that bare nouns can be either weak or strong definites in Mandarin. Saha et al. (2024) suggest that Mandarin actually does not distinguish weak from strong definites at all. To test the various hypotheses at play here, we turn in the next two sections to two manipulations of the basic donkey-anaphora setup which can in fact distinguish weak from strong definites.

⁸For Guébie, the question is whether a certain example features a donkey reading or an existentially quantified reading—a thorny question in other cases as well (cp. Chierchia 1992: §2).

⁹Moroney (2021a,b) discusses several additional languages where potentially weak definites can be donkey anaphors: Shan, Serbian, Kannada, and Thai. For her, donkey anaphora is a firm test for strong definiteness, and thus she takes these data to show that the relevant forms are weak/strong ambiguous. Our conclusion that donkey anaphora is *not* a test for strong definiteness makes possible an analysis where such forms are unambiguous weak definites after all.

3 No proper binder: predictions from dynamic semantics

Consider the ways that donkey readings arise for strong definites. Schwarz (2009) argues that strong definites contain indices, i.e. individual variables, as shown in (2), and that these indices are subject to dynamic binding. In an example like (11), on the hypothesis that the demonstrative DP *na mei jiangpai* ‘that medal’ is a strong definite, its binder is the indefinite *yi mei aoyun jiangpai* ‘an Olympic medal’ inside the restrictor of the quantificational subject. (Following Barwise (1987), we superscript binder indices and subscript anaphoric indices.) On dynamic theories, it is the distinctive dynamic nature of existential quantifiers that allows them to bind variables outside their c-command domain in this way.¹⁰

- (11) *mei*⁶ *ge* *na-guo* *yi*⁷ *mei aoyun jiangpai de yundongyuan dou ba na*₇
 every CLF take-EXP one CLF Olympics medal REL athlete DOU BA DEM
mei jiangpai *haohao zhencang qilai-le*
 CLF medal well cherish RES-ASP
 ‘Every⁶ athlete who won an⁷ Olympic medal cherished that₇ medal well.’

An important prediction for dynamic theories is that binding is possible only when a proper antecedent—in this case, an existential quantifier—is available. This prediction has been prominently discussed in connection with the ‘formal link’ between a donkey pronoun and its antecedent (Heim 1982). As Grosz et al. (2015) note, the problem of the formal link is closely connected to what Postal (1969) called ‘anaphoric islands’: a word that merely implies the existence of an individual (e.g. *married*) generally cannot antecede anaphora to that individual, as in (12), and neither can a sub-part of another word, as in (13).¹¹

- (12) a. Every man who has a¹ wife is sitting next to her₁. (Heim 1982)
 b. Every married man is sitting next to her. (no covarying reading)
 (13) a. Every man who owns a² donkey loves it₂. (Patel-Grosz and Grosz 2010)
 b. Every donkey-owner loves it. (no covarying reading)

Data like (12) and (13) have been discussed extensively from the perspective of theories that adopt only dynamic binding mechanisms for pronouns and definites (Heim 1982) as well as those that adopt only non-dynamic mechanisms (e.g. Elbourne 2001, Patel-Grosz and Grosz 2010). This discussion has focused on examples where the donkey anaphor is pronominal, and indeed the pronominality of the donkey anaphor has sometimes been taken as central to the analysis (Elbourne 2001, 2005).¹² We suggest that the manipulation in

¹⁰For an introduction to dynamic approaches to donkey anaphora, see Brasoveanu and Dotlačil (2020).

¹¹See however Patel-Grosz and Grosz (2010) and Grosz et al. (2015) for discussion of counter-examples involving pronouns in English, along with analyses of English pronouns that respond to these findings; see Patel-Grosz and Grosz (2017) for an alternative take on these data (along with German data). As our focus here is on full definite expressions rather than pronouns, we set these issues aside.

¹²There has also been some discussion from the perspective of ‘mixed’ dynamic/situation theories, again focusing on pronominal cases. See Chierchia (1992), Patel-Grosz and Grosz (2017).

question also has important, though less recognized, diagnostic value as applied to donkey anaphora with full definite description donkey anaphors. From the perspective of a theory of definites that includes both weak and strong varieties, along with dynamic and non-dynamic routes to covariation in donkey sentences, the manipulation of the antecedent in structures like (12) and (13) is expected to have an asymmetrical effect on strong definite donkeys vs. weak definite donkeys. Strong definite donkeys depend on dynamic binding and so should have a covarying reading *only when a proper antecedent is available*. But weak definite donkeys (and ambiguous forms that can be parsed as weak definites) should be able to have a covarying reading even without any such antecedent.

This prediction is borne out in German. In the baseline sentence in (14), the antecedent is the full indefinite *ein Haus* ‘a house’, and both weak and strong donkey definites are possible. The test case is (15), where *Haus* occurs only as a subpart of the word *Hausbesitzer* ‘homeowner’. With this manipulation, the strong definite can no longer express the co-varying reading, though the weak definite remains able to do so.¹³

- (14) Jeder, der ein Haus besitzt, hat ein Bild { vom / von dem }
 every that a house owns, has a picture of.the.WEAK of the.STRONG
Haus auf seinem Telefon
 house on 3SG phone
 Everyone who owns a house has a picture of the house on their phone.
- (15) Jeder Haus-besitzer hat ein Bild { vom / #von dem } **Haus**
 every house-owner has a picture of.the.WEAK of the.STRONG house
 auf seinem Telefon
 on 3SG phone
 Every home-owner has a picture of the home on their phone.

Given the logic we have outlined, this contrast is entirely as predicted by Schwarz (2009).¹⁴ With the full indefinite *ein Haus* ‘a house’ available as a dynamic binder, the strong definite is possible in (14). Removal of the dynamic binder in (15) takes this possibility away. The weak definite lacks an index and thus is not dependent on dynamic binding. Accordingly, it is not sensitive to this manipulation.¹⁵

We now turn to English. Following Schwarz (2009), the standard view on English definites in the literature is an ambiguity view: *the* may instantiate either the weak definite structure or the strong definite structure (see e.g. Jenks 2018, Moroney 2021a,b, Dayal and Jiang 2022). The literature also typically assumes that the English demonstrative *that* is strong in the sense that it contains an anaphoric index. The proposed ambiguity in this system makes it useful to clarify the logic of the no-proper-binder test. As noted in (16a), the mere possibility of a weak definite parse is enough for *the*-donkeys to be possible as donkey anaphors in no-proper-binder configurations. Conversely, if the donkey reading is

¹³Thanks to Florian Schwarz (p.c.) for these examples.

¹⁴It also tracks findings on German pronouns discussed by Patel-Grosz and Grosz (2010, 2017).

¹⁵Rather, the availability of the weak definite should vary with the situational uniqueness of a referent; see discussion in Grosz et al. (2015). We return to uniqueness manipulations in the next section.

ruled out for demonstratives in these contexts (as we show below), that provides evidence that demonstratives can only be strong definites, (16b).

(16) No-proper-binder test

- a. If a definite expression DX can be a donkey anaphor without a proper antecedent, DX can be a weak definite.
- b. If a definite expression DX can't be a donkey anaphor without a proper antecedent, DX is a strong definite.

What the test cannot do is confirm whether *the* also has a life as a strong definite. The results for English: *the* can be a weak definite and *that* cannot be. In contrast to the baseline (a) examples, the (b) examples below lack a proper antecedent. Here, donkey anaphora remains possible with *the* but not with *that*.¹⁶

- (17) a. Every person in this building who owns a dog already suspects that **the/that dog** is bothering the neighbors.
- b. Every dog-owner in this building already suspects that **the/#that dog** is bothering the neighbors.
- (18) Many people have the goal of writing a memoir, but few succeed.
 - a. Most people who successfully write a memoir keep **the/that memoir** focused on their experiences in adult life.
 - b. Most successful memoir-writers keep **the/#that memoir** focused on their experiences in adult life.
- (19) a. Unfortunately, no one who won a prize mentioned **the/that prize** to their spouse.
- b. Unfortunately, no prize-winner mentioned **the/#that prize** to their spouse.

We return to the question of how strong definiteness is expressed in English in section 4.

This brings us back to the contested matter of definiteness types in Mandarin. As we will show, the no-proper-binder test reveals a distinction: bare nouns can be weak definites, and demonstrative DPs cannot be. Returning to (11), assuming the demonstrative is a strong definite, we expect that the co-varying reading of the demonstrative DP will be lost when 'medal' is part of a compound, rather than a full indefinite DP. This prediction is borne out. Assuming in a parallel way that the bare noun is (Jenks 2018) or can be (Moroney 2021a,b, Bremmers et al. 2022, Dayal and Jiang 2022) a weak definite, and that a weak definite does not require a full indefinite antecedent, we expect that the bare noun will retain the co-varying reading when the antecedent is incorporated. This prediction is borne out, too.

¹⁶These judgments are those of the first author and three additional native speakers of US English.

- (20) mei ge aoyun **jiangpai-dezhu** dou ba (#na mei) **jiangpai** haohao
 every CLF Olympics medal-recipient DOU BA DEM CLF medal well
 zhencang qilai-le
 cherish RES-ASP
 ‘Every Olympic medalist cherished the medal well.’

Before turning to further examples, we note that many Mandarin words have both a monosyllabic and a disyllabic form. The disyllabic form (e.g. *fangzi* ‘house’) freely occurs in argument positions, while the monosyllabic form (here *fang* ‘house’) only occurs incorporated into compounds or conventionalized/idiomatic expressions. In (21a), with full indefinite antecedent *yi tao fangzi* ‘a house’, both a demonstrative DP and a bare noun can express the co-varying reading. In (21b), with incorporation structure *fang-nu* ‘mortgage slave’ (i.e., mortgage-burdened home-owner), the demonstrative DP can no longer co-vary.

- (21) a. dabufen wei-le yi tao **fangzi** beifu da’e daikuan de ren dou yao
 most for-ASP one CLF house shoulder big loan REL person DOU will
 yong gongzi de baifenzhi qishi gei (na tao) **fangzi** huan fangdai
 use salary REL percent seventy for DEM CLF house repay mortgage
 ‘Most people who are burdened with a big loan for a house will use seventy-percent of their income to pay for the mortgage of that house.’
 b. dabufen **fang-nu** dou yao yong gongzi de baifenzhi qishi gei (#na
 most house-slave DOU will use salary REL percent seventy for DEM
tao) fangzi huan fangdai
 CLF house repay mortgage
 ‘Most mortgage slaves will use seventy-percent of their income to pay for the mortgage of the house.’

The same pattern is shown in (22) for standalone *banji* vs. incorporated *ban* ‘class’.¹⁷

- (22) a. bulun shui dangxuan chengwei yi ge **banji** de ganbu, ta dou hui
 no.matter who be.elected become one CLF class REL officer 3SG DOU will
 dingqi wei (na ge) **banji** chouji ban-fei
 regularly for DEM CLF class raise class-fee
 ‘No matter who is elected the representative for the class (student-teacher liaison), they will have to regularly raise funding for that class.’
 b. bulun shui dangxuan chengwei **ban-ganbu**, ta dou hui dingqi wei
 no.matter who be.elected become class-officer 3SG DOU will regularly for
 (#na ge) **banji** chouji ban-fei
 DEM CLF class raise class-fee
 ‘No matter who is elected class-representative (student-teacher liaison), they will have to regularly raise funding for the class.’

¹⁷Note that (21) is a relative-clause type donkey sentence and (22) is a *dou*-conditional.

These findings provide new evidence that Mandarin indeed distinguishes weak and strong definites (Jenks 2018, Moroney 2021a,b, Dayal and Jiang 2022, Jenks and Chen 2026, *contra* Saha et al. 2024). In view of ambiguity theories, what they show most clearly is that bare nouns *can* be weak definites and that demonstrative DPs *cannot* be.

We note that a small complication of the no-proper-binder test for Mandarin comes from the findings of Ahn and Zhu (2025), who discuss Mandarin demonstratives in bridging contexts such as (23). Here, while there is no indefinite ranging over people in the quantifier restriction that can be a dynamic binder for the strong definite *na wei xiaoshua-jia* ‘that novelist’, there is an indefinite quantifier ranging over novels.

- (23) Wang Yawen xihuan shoucang tushu. ta mei-ci zhao-dao le yi ben
 Wang Yawen like collect book she every-time find-arrive ASP one CLF
 xihuan de **xiaoshuo**, zuihou dou hui faxian ziji du-guo **na wei**
 like REL novel finally always will discover self read-EXP DEM CLF
xiaoshuo-jia xie de ling yi ge gushi
 novel-person write REL another one CLF story
 ‘Wang Yawen likes to collect books. Every time she finds a novel that she likes,
 she eventually realizes that she has read another story written by the novelist.’ (Ahn
 and Zhu 2025: 472)

To capture this type of data (which, as they show experimentally, is quite different from what occurs with demonstratives in English), Ahn and Zhu (2025) propose a minor modification of the semantics for strong definites in (2): the relation between the referent of the index and the referent of the DP is contextually provided, and need not be identity. (This requires proposing a slight difference in denotation for the Mandarin demonstrative vs. English *that*. We return to this idea in section 5.)

- (24) $\llbracket na\ wei\ xiaoshua-jia\ 7 \rrbracket^g = \text{‘the unique } x \text{ such that } x \text{ is a novelist and stands in some contextually-given relation to } g(7)\text{’}$ (after Ahn and Zhu 2025: 468)

On this view, the donkey demonstrative *na wei xiaoshua-jia* ‘that novelist’ has a proper antecedent as its binder; the dynamic binder is *yi ben xihuan de xiaoshuo* ‘a novel that she likes’. Thus (23) is not a counterexample to our claim that Mandarin donkey demonstratives require a proper antecedent. Rather, it shows that this antecedent may stand in a bridging relation, rather than a relation of identity, to the Mandarin strong definite.

This approach raises a potential challenge for our account of examples like (20), repeated below with one possible pattern of dynamic indexing. It remains the case that there is no dynamic antecedent ranging over medals, but there is an antecedent ranging over people related to medals in a contextually salient way. If an analysis like (26) is available for the demonstrative DP, why do we not see a covarying reading for *na mei jiangpai* ‘that medal’ in this example in the way that we do for *na wei xiaoshuo-jia* ‘that novelist’ in (23)?

- (25) # mei³ ge aoyun jiangpai-dezhu dou ba **na₃** **mei jiangpai** haohao
 every CLF Olympics medal-recipient DOU BA DEM CLF medal well
 zhencang qilai-le
 cherish RES-ASP
 Intended: ‘Every Olympic medalist cherished the medal well.’
- (26) $\llbracket na\ mei\ jiangpai\ 3 \rrbracket^g =$ ‘the unique x such that x is a medal and stands in some contextually given relation to $g(3)$ ’

We suggest that the relevant factor here is a constraint on demonstratives we also see in other languages, e.g. German (Hinterwimmer and Brocher 2018): Mandarin demonstratives resist binding by a subject quantifier. In Ahn and Zhu’s example (23), the quantifier over novels is not in subject position. When we modify the structure to place the relevant quantifier in subject position, the demonstrative is no longer available.

- (27) # mei² ben changxiao xiaoshuo dou hui wei **na₂** **wei xiaoshuo-jia** dailai
 every CLF best-selling novel DOU will for DEM CLF novel-person bring
 feichang keguan de shouru
 very considerable REL income
 ‘Every best-selling novel would bring in considerable income for the novelist.’

Methodologically, what we take away here is that for anaphoric expressions that allow bridging (e.g. Mandarin demonstratives, but not English demonstratives), no-proper-binder test cases must be carefully constructed so as to rule out alternative antecedents other than the incorporated one. The absence of a co-varying reading in (21b), (22b), and (25) arises because the ‘ordinary’ antecedent for the strong definite is not a proper quantifier, *and* the only other quantifier in the sentence is in an inadmissible syntactic position.

4 Bishop sentences: predictions from situation semantics

Let us now consider the ways that donkey readings arise for weak definites. Weak definites are simple situation-relative uniqueness operators: (when defined,) given a situation s , they return the unique member of the NP denotation in s . Covarying readings of weak definites are possible in donkey sentences because these sentences involve quantification over situations (Heim 1990, Elbourne 2005, Schwarz 2009, i.a.). In (28), we show how this analysis applies to an example repeated from above (slightly simplifying from Schwarz 2009).

- (28) a. mei ge you yi ming latuoweiya qiuyuan de bingqiu dui dou hui
 every CLF have one CLF Latvian player REL hockey team DOU will
 rang **latuoweiya qiuyuan** dang shoumenyuan
 let Latvian player serve.as goalie
 ‘Every hockey team that has one Latvian player would make the Latvian player the goalie’

- b. For every individual x and situation s_2 , if s_2 exemplifies the proposition that x is a hockey team with a Latvian player, then there exists a situation s_3 which extends s_2 and x makes the unique Latvian player in s_3 be the goalie in s_3 .

A crucial notion here is *exemplification*, a form of minimality; as Kratzer (2017) writes, “intuitively, a situation that exemplifies a proposition p is one that does not contain anything that does not contribute to the truth of p .” Given this, in (28), we are concerned only with values of s_2 that contain a hockey team with a Latvian player *and nothing else*. The sentence requires for its truth that all such situations can be extended to larger situations in which there is a unique Latvian player (and that player is made goalie).

The role of uniqueness in this analysis predicts a limitation on where weak donkey definites can be used. The definite is evaluated relative to situations that extend some minimal (proposition-exemplifying) situation determined by the restrictor of the quantifier. What if those starting, minimal situations already contain multiple individuals that meet the NP description? The weak definite should be impossible. This prediction is borne out in German, as Schwarz (2009) discusses. Sentence (29) involves quantification over situations that involve multiple ministers. In the main clause, a weak definite *beim Minister* ‘by the Minister’ is impossible. (In contrast, a weak definite picking out a referent that remains (situation-relatively) unique, such as *beim Kanzler* ‘by the Chancellor’, remains possible.)

- (29) Wenn ein Minister den anderen Ministern im Kabinett den Haushalt
 when a minister the other ministers in.the.WEAK cabinet the budget
 kürzt, dann gehen **beim** { #Minister / Kanzler } viele Beschwerden
 cuts then go by.the.WEAK minister chancellor many complaints
 ein
 in
 When a minister cuts the budget on other ministers in the cabinet, that minister /
 the chancellor receives a lot of complaints. (Schwarz 2009: 245)

As Schwarz (2009), Coppock (2022), and others have discussed, the relevant examples closely mirror those discussed in the pronouns literature as ‘bishop’ sentences, e.g. (30).¹⁸

- (30) When a bishop meets a bishop, he blesses him.

Such examples pose what Heim (1990: 148) dubbed ‘the problem of indistinguishable participants’; there is not a unique referent available for a definite description (e.g. *the bishop*). We will refer generally to examples that pose the problem of indistinguishable participants, frustrating the use of uniqueness (weak) definites, as bishop sentences.

Bishop sentences provide a test for strong definiteness, as Schwarz (2009) and Coppock (2022) observe. Keeping in mind that some forms may be ambiguous, as discussed in the

¹⁸While the analysis of pronouns is not our topic here, we note that (30) is a classic problem for theories of pronouns that analyze them as akin to weak definite descriptions (see e.g. Kratzer 1989: 66, Heim 1990). The well-formedness of (30) could thus be taken as evidence that English pronouns have a source that is not a weak definite. See Patel-Grosz and Grosz (2017) for further discussion of pronoun syntax and semantics.

previous section, the fact that a definite expression can be used as a donkey anaphor in a bishop sentence provides evidence that it has a parse as a strong definite, (31a). Only an unambiguous weak definite form is predicted to be ruled out in a bishop context, (31b).

(31) Bishop test

- a. If a definite expression DX can be a donkey anaphor in contexts of indistinguishable participants, DX can be a strong definite.
- b. If a definite expression DX can't be a donkey anaphor in contexts of indistinguishable participants, DX is a weak definite.

For Mandarin, this test provides a means to compare the view that bare nouns are weak/strong ambiguous (Moroney 2021a,b, Bremmers et al. 2022, Dayal and Jiang 2022) with the view that bare nouns are always weak (Jenks 2018, Jenks and Chen 2026). On the ambiguity view, we expect that bare nouns will be possible donkey anaphors in bishop contexts. On the strictly-weak view, we expect not. The evidence favors the strictly-weak view. Consider first a version of (28) without the repeated adjective *latuoweiya* 'Latvian'. In this case, if the donkey anaphor is a weak definite, the sentence has the truth-conditions in (32b). We are concerned with situations exemplifying the proposition that *x* is a hockey team with a Latvian player. Such situations include those with multiple players. But (32b) requires extensions of all such situations to include exactly *one* player. This cannot be.

- (32) a. mei ge you yi ming latuoweiya qiuyuan de bingqiu dui dou hui
 every CLF have one CLF Latvian player REL hockey team DOU will
 rang #(na ming) qiuyuan dang shoumenyuan
 let DEM CLF player serve.as goalie
 'Every hockey team that has a Latvian player would make that player the goalie'
- b. # For every individual *x* and situation *s*₂, if *s*₂ exemplifies the proposition that *x* is a hockey team with a Latvian player, then there exists a situation *s*₃ which extends *s*₂ and *x* makes the unique player in *s*₃ be the goalie in *s*₃.

Turning to the form of the donkey anaphor, we see in (32a) that a bare noun cannot be used; only the demonstrative is acceptable. By the logic of (31), this fact provides evidence that the bare noun *qiuyuan* 'player' has no parse as a strong definite, *contra* Moroney (2021a,b), Bremmers et al. (2022), Dayal and Jiang (2022). The only form that can be a strong definite is the demonstrative DP. Further examples in support of this generalization are provided in (33) and (34). Example (33) makes use of the same logic as (32), but with car shows including multiple cars. Example (34) is intended to parallel Schwarz's example (29).

- (33) mei ge you yi liang asidunmading de chezhan dou hui ba #(na liang)
 every CLF have one CLF Aston.Martin REL car.show DOU will BA DEM CLF
che fang zai menkou zui xianyan de difang
 car put at entrance most eye-catching REL place
 ‘Every car show that has an Aston Martin would put that car at the most eye-catching spot near the entrance.’
- (34) ruguo yi wei guyuan zongshi jiang qita guyuan de bushi, name
 if one CLF employee always tell other employee REL wrongdoing then
 women bixu kaichu #(na wei) guyuan
 1PL must fire DEM CLF employee
 ‘If an employee is always criticizing other employees, we must fire that employee.’

Before we move to English bishop examples, we note that a potential alternative analysis of this type of data might draw on the observation that demonstratives sometimes imply ‘anti-uniqueness’: *that F* requires that there be an additional element in the domain that is F, besides the referent (Robinson 2005). For instance, examples like (35) are ill-formed, unless multiple suns are under discussion; similar facts hold in Mandarin.

- (35) #**That sun** is made of hydrogen and helium (Robinson 2005: 51)

Nowak (2021), Dayal and Jiang (2022), and Saha et al. (2023) propose that antiuniqueness is encoded in demonstratives as a presupposition; this would mean that *na wei guyuan* ‘that employee’ presupposes that there is more than one salient employee. Suppose this is so and suppose furthermore that strong definites are different from demonstratives and have a logically weaker presupposition (Nowak 2021, Saha et al. 2023). Then the choice of a demonstrative in (32)-(34) could be modeled as a Maximize Presupposition effect (Heim 1991). This makes room for an ambiguity analysis for bare nouns to survive after all: the bare noun *can* be a strong definite, but the demonstrative outcompetes it in bishop contexts on account of its stronger presupposition.

The attractiveness of this type of counteranalysis depends on the attractiveness of hardwiring the anti-uniqueness implication in demonstratives as a presupposition. Blumberg (2020) and Ahn (2022, 2025) argue that this kind of approach misses a crucial generalization about when demonstratives imply antiuniqueness: precisely in those contexts where they compete with definites. In an example like (35), the demonstrative implies antiuniqueness because the uniqueness meaning could have been expressed in a more optimal way by the use of the weak definite. Deictic demonstratives do not compete with the weak definite and therefore show no antiuniqueness effect, (36). Ahn (2022) argues that the same holds for cases like (37) where a relative clause is present.

- (36) I don’t know if there are any other cars in this showroom, but [pointing to a particular car] **that car** looks expensive. (Blumberg 2020: 49)
- (37) **That hominid** who discovered how to start fires was a genius. (King 2001: 9)

Blumberg and Ahn conclude that antiuniqueness effect in (35) is an implicature, arising pragmatically from the use of a more complex form (a demonstrative) when a simpler one (a weak definite) is available. In view of these arguments, we do not see a way to maintain the ambiguity theory about Mandarin bare nouns in bishop sentences. Demonstratives are required instead of bare nouns in (32)-(34) not because they have a (desirably) strong presupposition, but because only demonstratives can be strong definites. Bare nouns are only weak definites, inadmissible in bishop sentences on account of their uniqueness semantics.

Coming now to English bishop sentences, we are again in a position to probe a proposed ambiguity; the standard view of English *the* is that it is weak/strong ambiguous. The no-proper-binder test provided positive evidence for a weak definiteness analysis. The bishop test indicates that *the* has only a weak analysis. In (38), the donkey definite may use *that* but not *the*. By (31), this indicates that *the* does not have a parse as a strong definite.^{19 20}

- (38) a. Every hockey team that has a Latvian player makes **that/#the player** the goalie.
 b. Every group of books that contains a book about Warhol is displayed with **that/#the book** on top.
 c. If an employee is criticizing other employees, we must fire **that/#the employee**.

We conclude that neither English nor Mandarin has an ambiguous weak/strong definite form. English *the* and Mandarin bare nouns are strictly weak definites. Strong definiteness contexts call for the use of the demonstrative in both languages.

¹⁹We have confirmed the judgments in (38) with five native speakers of US English. As for the Mandarin data, it is of course possible that different varieties or even idiolects of English vary regarding these judgments. One additional US English speaker we consulted allowed *the* and *that* equally in all cases, which could indicate that this individual has a different grammar of definiteness. Perhaps relatedly, the few works that have discussed examples like (38) report varying levels of degradation for the *the* versions (e.g. ‘none of these are particularly good’, Schwarz 2009: 244; ‘slightly stilted’, Elbourne 2001: 268). See below for further discussion of an example that our consultants did not all agree on (the English translation of German (40)). English bishop sentences call out for assessment in a controlled experimental setting.

²⁰Note that our English bishop sentences diverge in two ways from those discussed by Coppock (2022). First, ours lack main-clause ‘other’ (cf. *If a bishop meets a bishop, then the bishop blesses the other bishop*), which we take to make the two participants distinguishable after all. Second, while Coppock discusses (i)-(ii), with *the*, as English versions of German examples, it is not clear to us whether she means to present any particular judgments on these sentences. Various English speakers we have consulted find *the* worse than *that* in both cases, though judgments are somewhat mixed (further cause for proper experimental assessment).

- (i) When a minister cuts the budget of other ministers in the cabinet, **that/??the minister** receives a lot of complaints.
 (ii) When a professor recommends a student to another professor, his application is read by **that/??the professor** with great attention.

5 Conclusions and further directions

Different technologies for definiteness predict different distributions for definite expressions. Our main goal in this paper has been to better understand contexts for weak and strong definites on the grounds of the underlying technology, focusing on donkey anaphora.²¹ Our main conclusions are three. One, donkey anaphors can be weak or strong definites. Two, when there is no proper antecedent, only weak definites can be donkey anaphors. Three, in a situation of indistinguishable participants, only strong definites can be donkey anaphors. We have shown how the three conclusions apply in the weak/strong definiteness system of German, expanding on Schwarz's (2009) proposed diagnostics by including no-proper-binder type cases. We have also shown how the three conclusions apply to Mandarin, where weak/strong diagnostics have been contested, and to English. Given our tests, we have diagnosed Mandarin bare nouns and English *the* both as strictly weak.

This conclusion diverges from much literature on English (e.g. Schwarz 2009, Moroney 2021a,b, Dayal and Jiang 2022) and Mandarin (e.g. Moroney 2021a,b, Dayal and Jiang 2022, Saha et al. 2024). We suggest that this may reflect differences between German, English, and Mandarin that are in need of much greater study. Consider for instance simple (non-donkey) anaphoric uses. Both in German and Mandarin, the weak definite form can be used anaphorically in certain cases, but subject to restrictions that remain incompletely understood. For Mandarin, much recent literature has discussed these restrictions (see Jenks 2018, Bremmers et al. 2022, Dayal and Jiang 2022, Simpson and Wu 2022); details of the German situation are discussed in Schwarz (2009: 43-49). English *the*, in contrast, is broadly licit in anaphoric uses and not sensitive to the factors that matter in German or Mandarin. In (39), for instance, Schwarz reports that German requires the strong article, but English speakers allow weak *the*, as shown in the free translation:

- (39) In der New Yorker Bibliothek gibt es ein Buch über Topinambur. Neulich
 in the New York library exists EXPL a book about topinambur recently
 war ich dort und habe { **#im** / **in dem** } **Buch** nach einer
 was I there and have in.the.WEAK in the.STRONG book for an
 Antwort auf die Frage gesucht, ob man Topinambur grillen kann.
 answer to the question searched whether one topinambur grill can
 'In the NY public library, there is a book about topinambur. Recently, I was there
 and searched in **the book** for an answer to the question of whether one can grill
 topinambur.' (Schwarz 2009: 30)

Uniqueness theories account for anaphoric uses by means of domain restriction. In (39), as these theories would have it, after the first sentence, a situation containing a single book is salient; this situation is used in the calculation of uniqueness for the following sentence.

²¹A future direction concerns existential vs. universal readings of donkey anaphors (see e.g. Brasoveanu and Dotlačil 2020). As Chierchia (1992: 163) notes, weak definites should have strictly universal readings, whereas strong definites have existential readings. (See also Schwarz 2009: 178-9 for partial agreement with Chierchia, and Cheng and Huang 1996 for a dissenting view.) We have not attempted to test this prediction.

The English anaphora facts are no problem, then, on an off-the-shelf theory of *the* as weak. As Coppock (2022: 129) notes, it remains unclear why the weak article cannot be similarly used in German. We find a similar puzzle in Mandarin. Given that weak and strong definites have an overlapping distribution in both languages, we do not think facts like (39) can be accounted for via a competition principle favoring the strong definite (e.g. Jenks' (2018) *Index!*).²² Rather, the solution seems to lie in a requirement in these languages that a slightly larger domain be taken into consideration (at least sometimes) for weak definites.

Similar remarks apply to outstanding questions of variation in donkey anaphora. In (40), the German weak article is again ruled out according to Schwarz, but (for three of the five English speakers we consulted) *the* is acceptable in the English translation.

- (40) In jeder Bibliothek, die ein Buch über Topinambur hat, sehe ich { **#im**
in every library that a book about topinambur has look I in.the.WEAK
/ **in dem** } **Buch** nach, ob man Topinambur grillen kann.
in the.STRONG book PART whether one topinambur grill can
'In every library that has a book about topinambur I check in **that/%the book**
whether one can grill topinambur.' (Schwarz 2009: 242)
- (41) For all individuals *x* and situations *s*₂, if *s*₂ exemplifies the proposition that *x* is a library that has a book about topinambur, then there exists a situation *s*₃ which extends *s*₂ and in *s*₃, in *x*, I check in the unique book in *s*₃ whether one can grill topinambur.

Contrasts like this could indicate that English *the* is not strictly a weak article, as Schwarz suggests. But this leaves the judgment on the otherwise similar examples in (38) mysterious. Particularly notable is the contrast between (40) with *library* and (38b) with *group of books*. In our discussion of *team*, we assumed that situations that exemplify a team may include a plurality of players. The same presumably applies to *group of books*. *Library* behaves differently: we seem to need exemplification to rule out situations with a plurality of books (thus making it possible for extensions of all such situations to contain a single book). A natural hypothesis is that *library* is understood as describing buildings, not book-collections, in (40). This would mean that the situations quantified over in (41) contain only building-related properties (but no books)—and that English *the* behaves as predicted for a uniqueness definite, whereas the German weak article does not. In particular, if our reasoning applies for *Bibliothek* 'library', it seems the weak article in German again requires uniqueness relative to a larger situation than the off-the-shelf semantics predicts.

What final view of the typology of definiteness does this leave us with? On one hand, as we have discussed at length in the last two sections, we do find evidence for distinct uniqueness- and familiarity-based definiteness technologies at work in natural language, with a relatively stable profile in our test environments across the three languages we have discussed. On the other hand, it seems unlikely in view of the considerations just discussed

²²In rejecting *Index!*, we join Simpson and Wu (2022), Bremmers et al. (2022), Dayal and Jiang (2022), Zheng and AnderBois (2023), Ahn and Zhu (2025).

that all uniqueness-based definites across languages have a totally identical semantics, or that all familiarity-based definites do. More likely is that weak definiteness across languages represents a cluster of possible uniqueness-based definite meanings and that strong definiteness represents a cluster of possible familiarity-based meanings, as already suggested e.g. by the work of Owusu (2022) or Agodio et al. (2024). This kind of conclusion finds support in Ahn and Zhu’s (2025) comparative study of bridging anaphora in English and Mandarin, which finds that Mandarin demonstratives allow bridging where English demonstratives do not. To capture this, as noted above, Ahn and Zhu propose a minor difference in the semantics of Mandarin demonstrative *na* versus the strong definite in (2) (repeated below in (42a)). Both take a situation argument, a predicate, and a variable, but where the strong definite required identity between the referent and the value of the variable, *na* requires only relatedness via a contextually-given relation (R^c), (42b).

- (42) a. $\llbracket \text{DEF}_{\text{strong}} \rrbracket^{g,c} = \lambda s. \lambda P. \lambda y. \iota x [P(x)(s) \wedge x = y]$ (Schwarz 2009: 264)
 b. $\llbracket na \rrbracket^{g,c} = \lambda s. \lambda P. \lambda z. \iota x [P(x)(s) \wedge R^c(z, x)(s)]$ (Ahn and Zhu 2025: 466)

From our perspective, both meanings in (42) fall under the broad heading of strong definiteness, i.e. definiteness that involves individual variables. We expect to find further evidence for variation of this type within the broad classes of strong and of weak definites. We have suggested, for instance, that weak definites in English and German minorly differ regarding domain restriction, while both being variants of a uniqueness analysis; this may offer a solution to the puzzle of (40). It is our hope that the diagnostics we have put forward can provide solid ground for researchers looking to classify definites within the broadly strong and broadly weak categories on the way to finer-grained investigations of this sort.

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